



مدينة زويل للعلوم والتكنولوجيا
Zewail City of Science and Technology

REE 222 (Engineering Production) Project (Phase 1)

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Workpiece 1



Processes (sequenced)

- 1- Facing**
- 2- Straight turning**
- 3- Contouring**
- 4- External Threading**
- 5- Grooving**
- 6- Chamfering**

Tools

The following tools and processes are for the DMU 50 5-axis CNC:

Facing:

Tool: Facing tool (also known as a facing cutter)

Description: The facing tool is used to create a flat surface on the end face of the workpiece. It removes material to achieve a smooth finish.

Straight Turning:

Tool: Turning tool

Description: Straight turning involves removing excess material from the cylindrical surface of the workpiece to achieve the desired diameter. The turning tool moves parallel to the lathe axis.

Contouring:

Tool: Contouring tool

Description: contouring involves gradually changing the diameter of the workpiece along its length. The contouring tool creates conical surfaces.

External Threading:

Tool: Threading tool (also known as a threading insert)

Description: The threading tool is used to cut external threads on the workpiece. It follows a specific thread pitch and depth to create the desired thread profile.

Chamfering:

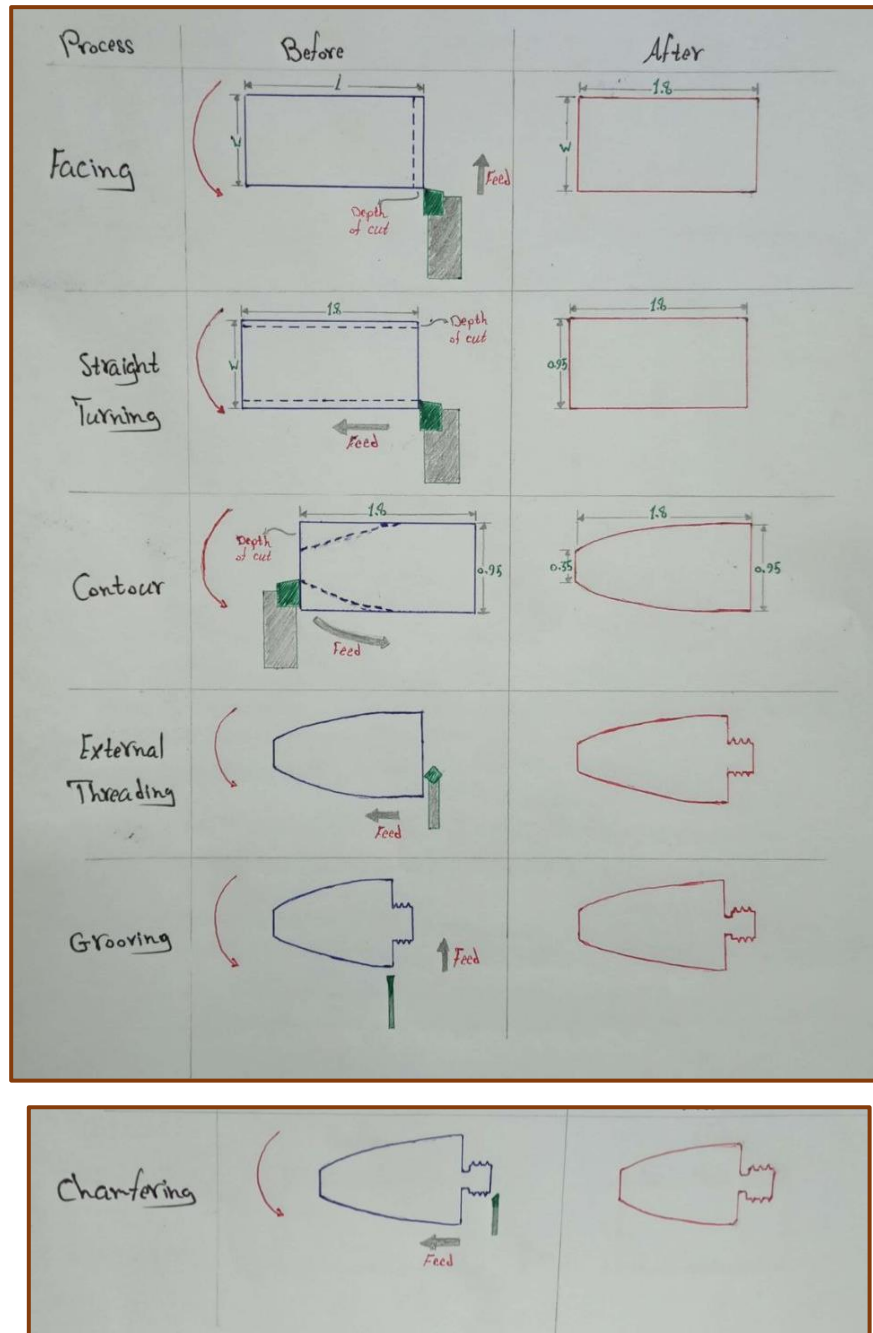
Tool: Chamfering tool

Description: Chamfering tools create beveled edges or chamfers on the workpiece. These beveled edges improve aesthetics, reduce sharp corners, and facilitate assembly.

Grooving:

Tool: Grooving tool (also known as a parting tool)

Description: The grooving tool cuts narrow grooves or slots into the workpiece. It is commonly used for parting off or separating the workpiece.



Calculations

Assuming typical spindle speed of 2650 rpm:

Facing:

- Facing time: 0.508 minutes
- The consistent feed = 0.1 inch/rev

Straight Turning:

- Turning time: 0.508 minutes
- Feed = 0.2 inch/rev

External Threading:

- Threading time: 0.4 minutes
- Feed = 0.068 inch/rev

Chamfering:

- Chamfering is a quick operation. Thus, let's assume a fixed time of 0.2 minutes.

Contour:

- Feed = 0.1 inch/rev
- Contouring time depends on the complexity of the shape. Assuming a simple contour, let's estimate 0.5 minute.

Grooving:

- Grooving time can vary. Let's assume a moderate time of 0.2 minute.

- Feed = 0.042 inch/rev

The total MRR = 0.792 inch³/rev

Workpiece 2



Processes (sequenced)

- 1- Facing**
- 2- Straight turning**
- 3- Grooving (half a circle shape)**
- 4- Internal Threading**
- 5- External Threading**
- 6- Grooving (straight)**
- 7- Chamfering**

Tools

The following tools and processes are for the DMU 50 5-axis CNC:

Facing:

Tool: Facing tool (also known as a facing cutter)

Description: The facing tool is used to create a flat surface on the end face of the workpiece. It removes material to achieve a smooth finish.

Straight Turning:

Tool: Turning tool

Description: Straight turning involves removing excess material from the cylindrical surface of the workpiece to achieve the desired diameter. The turning tool moves parallel to the lathe axis.

Internal threading:

Tool: Threading tool (also known as a threading insert)

Description: The tool cuts external threads on the workpiece. It follows a specific thread pitch and depth to create the desired thread profile.

External Threading:

Tool: Threading tool (also known as a threading insert)

Description: The tool cuts external threads on the workpiece. It follows a specific thread pitch and depth to create the desired thread profile.

Chamfering:

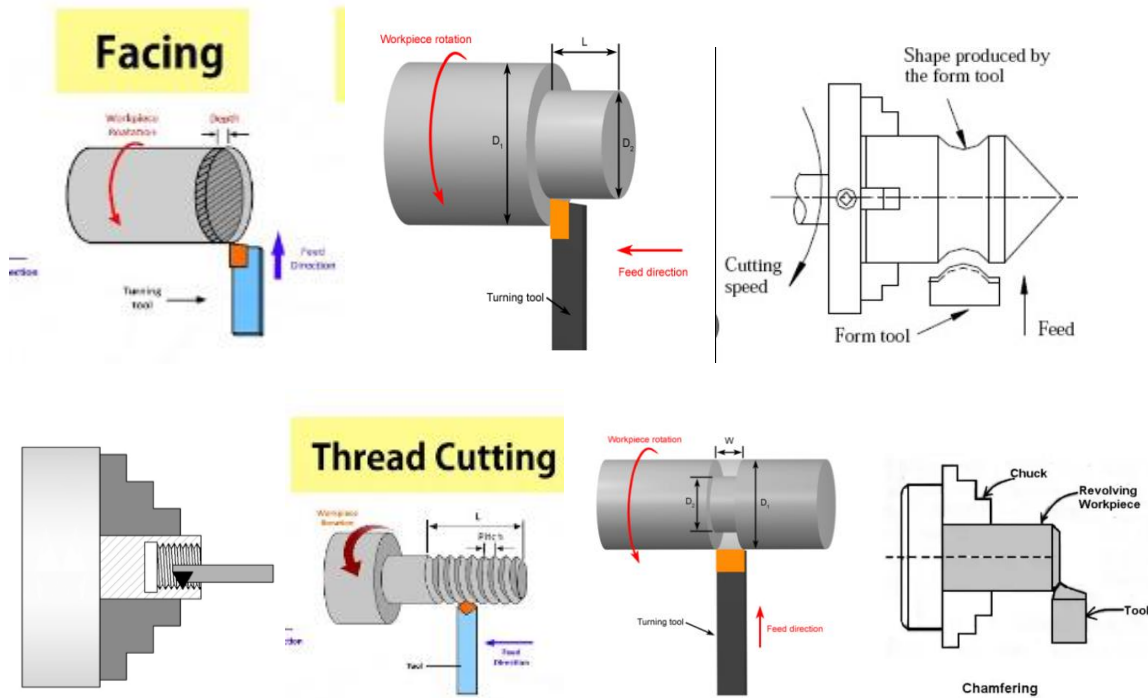
Tool: Chamfering tool

Description: Chamfering tools create beveled edges or chamfers on the workpiece. These beveled edges improve aesthetics, reduce sharp corners, and facilitate assembly.

Grooving:

Tool: Grooving tool (also known as a parting tool)

Description: The grooving tool cuts narrow grooves or slots into the workpiece. It is commonly used for parting off or separating the workpiece. The grooves may be straight or curved.



Calculations

Assuming typical spindle speed of 2650 rpm:

Facing:

- Facing time: 0.3. minutes
- The consistent feed = 0.165 inch/rev

Straight Turning:

- Turning time: 0.26 minutes
- Feed = 0.2 inch/rev

Internal Threading:

- Threading time: 0.5 minutes
- Feed = 0.2 inch/rev

Grooving (half circle):

- Chamfering is a quick operation. Thus, let's assume a fixed time of 0.4 minutes.
- Feed = 0.125

External threading:

- Feed = 0.03 inch/rev
- Contouring time depends on the complexity of the shape. Assuming a simple contour, let's estimate 0.5 minute.

Grooving:

- Grooving time can vary. Let's assume a moderate time of 0.3 minute.
- Feed = 0.0325 inch/rev

The total MRR = 0.682 inch³/rev